

METODOLOGY FOR GENERATING RANDOM VARIABLES

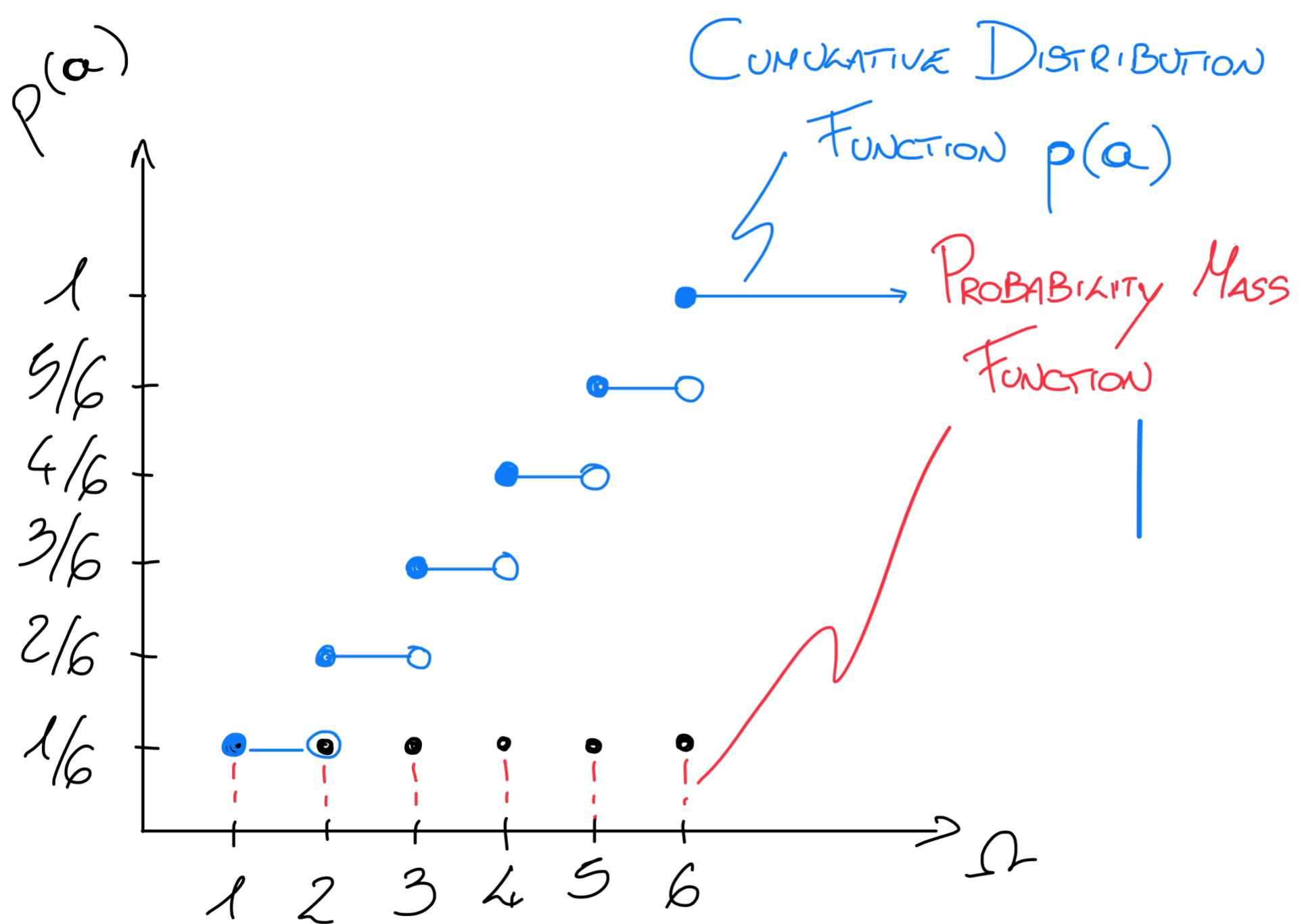
THE INVERSE TRANSFORM METHOD

- Discrete Random Variables
- Probability Mass Functions $\mapsto$ Map an item in the sample space in the corresponding distribution: $f: \Omega \rightarrow \mathbb{R}$

DICE ROLL

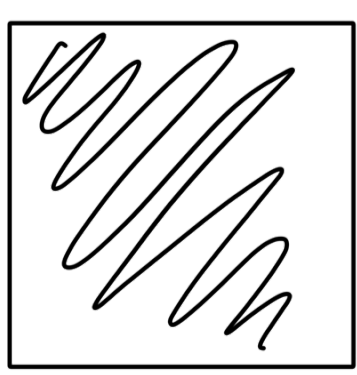
$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$p(a) = \begin{cases} 1/6, & a = 1, 2, 3, 4, 5, 6 \\ 0, & \text{otherwise} \end{cases}$$



$$\text{CDF } p(a) = \begin{cases} 0, & a \in (-\infty, 1) \\ 1/6, & a \in [1, 2) \\ 2/6, & a \in [2, 3) \\ 3/6, & a \in [3, 4) \\ 4/6, & a \in [4, 5) \\ 5/6, & a \in [5, 6) \\ 1, & a \in [6, \infty) \end{cases}$$

Extracting random values from $[0, 1]$



0.5 $\rightarrow$ 3
0.9 $\rightarrow$ 5
0 $\rightarrow$ 0
 $U \sim \text{Unif}(0, 1)$

For this example it's easy to use the inverse transform method

STEPS TO GENERATE A RANDOM VARIABLE

- 1) Find the CDF, $F(a)$
- 2) Invert the CDF, $G(a)$
- 3) Draw from $U \sim \text{Unif}(0, 1)$
- 4) Input + return $G(U)$